

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)
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Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

After reading some introduction of brown dwarf on Google, Brown dwarfs are substellar objects that are not massive enough to sustain stable hydrogen fusion. As they cool, condensate clouds can form in their atmospheres. Uneven cloud coverage and magnetic spots can cause brightness variations as the object rotates, making photometric variability a useful way to study brown dwarf atmospheres.

We propose time-series r- and i-band photometry of the nearby L3.5 brown dwarf 2MASSW J0036159+182110 using the Bok 90-inch telescope. This object is a known variable ultracool dwarf. Croll et al. (2016) measured a rotation period of 3.080 ± 0.001 hours and found wavelength-dependent variability, suggesting that starspots and atmospheric clouds may both contribute to its observed brightness variations.

Using Bok 90 telescope, we will try to monitor our target for around 8 hours on Sep 30th, covering about 2.6 rotation periods, and measure its flux relative to field stars to detect, or set a meaningful upper limit on, rotationally modulated variability.

Reference

Croll, B., Muirhead, P. S., Han, E., Dalba, P. A., Radigan, J., Morley, C. V., Lazarevic, M., & Taylor, B. 2016, "Long-term, Multiwavelength Light Curves of Ultra-cool Dwarfs: I. An Interplay of Starspots & Clouds Likely Drive the Variability of the L3.5 Dwarf 2MASS 0036+18," arXiv:1609.03586.

Telescope Requested (0.5 points): Bok 90-inch telescope

Instrument (0.5 points): 90 Prime

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

The brightness of moon will affect our observing process, since brown dwarf is pretty red so further from the moon, less impact from moon we can be received. Grey conditions are acceptable, although darker conditions are preferred, moonlight increases the sky background and reduces the signal to noise of our photometry, especially because our target is relatively faint, but differential photometry using nearby comparison stars should reduce some of these effects.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

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Object Name	Right Ascension (deg)	Declination (deg)	V magnitude (i-band and r-band)	Object Type	Visibility (on Sep 30 th)
2MASSW J0036159+182110	9.0671	18.3529	17.1922 19.5066	L3.5	~ 8.67h
2MASSW J0149090+295613	27.2875	29.9368	17.7745 19.9889	M9.5	~ 8h

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? (2 points)

Like what I wrote in abstract section, we need approximately 8 hours of continuous observing time. The known rotation period of 2MASSW J0036159+182110 is approximately 3.08 hours, so an 8-hour observation will cover about 2.5 complete rotations.

From Heinze et al. (2015), they used 200 s exposures with the F814W filter (0.7–0.95 μm), so we used their photometric precision as a reference. Their best precision was about 1.2% rms for a target with F814W ≈ 18.8 , which is roughly $S/N \approx 83$. Based on our target brightness, we estimate $S/N \approx 174$ in i and ≈ 60 in r. Therefore, an S/N range of about 50–200 should be reasonable for our observations.

Reference

Heinze, A. N., Metchev, S., & Kellogg, K. 2015, “Weather on Other Worlds. III. A Survey for T Dwarfs with High-Amplitude Optical Variability,” *The Astrophysical Journal*, 801, 104.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? (2 points)

The observations will provide r- and i-band light curves of 2MASSW J0036159+182110 over about 2.6 rotation periods. We will use differential photometry to search for repeating brightness variations and compare the variability between the two filters. This will allow us to characterize its rotational variability and compare our results with Croll et al. (2016).

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. (0.5 points)

If our primary target cannot be observed, we will observe our alternate target, 2MASSW J0149090+295613, an M9.5 dwarf that is visible for approximately 8 hours on Sep30th. We will use the same r- and i-band photometric monitoring strategy to search for brightness variability. If poor weather prevents observations entirely, we will use archival photometric data to study the variability of our primary target.